CRIMSON MAP: A Hybrid Augmented Reality Indoor-Outdoor Navigation System for University Campuses

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Abstract—CRIMSON MAP is a mobile-based augmented reality navigation system developed to improve campus wayfinding at Western Mindanao State University Main Campus. Large university campuses often present navigation challenges for students, visitors, and staff due to complex layouts and limited directional guidance. To address this problem, the system integrates Global Positioning System technology for outdoor navigation with Quick Response code scanning for indoor floor plan access. The application provides real-time route guidance, augmented reality directional overlays, audio prompts, and location tracking to assist users in navigating campus facilities efficiently. The system was developed using Unity and ARCore, incorporating an optimized A* pathfinding algorithm for route computation.

The system was evaluated through a structured software evaluation instrument involving thirty (n = 30) freshmen participants across three university campuses. Evaluation results indicate high levels of functionality, usability, reliability, and performance. The system achieved a mean usability score of 4.64 (SD = 0.42), reliability score of 4.53 (SD = 0.38), and overall satisfaction score of 4.56 (SD = 0.39) on a 5-point Likert scale. Quantitative performance metrics revealed an average localization error of 2.34 m (SD = 0.87 m) for GPS-based outdoor navigation and 0.42 m (SD = 0.15 m) for QR-assisted positioning, with route computation times averaging 0.18 s (SD = 0.04 s). The indoor floor plan feature received the highest rating at 4.80 (SD = 0.41), with 90% of participants rating it as 5. This study contributes a hybrid AR-GPS-QR navigation architecture with empirical validation across three university campuses, demonstrating that integrating augmented reality with location-based services significantly enhances navigation efficiency in large academic institutions.

Keywords—Augmented Reality, Mobile Navigation System, Campus wayfinding, GPS-based Navigation, QR code integration, location-based services, A* pathfinding.

I. Introduction

Navigating large university campuses presents significant challenges due to the presence of multiple buildings, offices, and facilities spread across wide areas. At Western Mindanao State University (WMSU) Main Campus, which comprises three geographically distributed campuses (A, B, and C) serving over 32,000 students, students and visitors frequently encounter difficulties locating specific destinations, particularly during initial campus visits. Conventional navigation aids such as printed maps and directional signages provide limited assistance and fail to offer real-time guidance, often resulting in confusion, delays, and missed appointments. This problem is further compounded by the absence of a unified digital wayfinding system that can adapt to the varying signal conditions across indoor and outdoor campus environments.

Recent developments in mobile technology have enabled the use of location-based services to improve navigation in complex environments. Augmented reality (AR) enhances user navigation by overlaying digital directional cues onto real-world views through mobile devices, offering more intuitive wayfinding support compared to traditional maps. Several studies have explored AR-based navigation systems for university campuses. Lu et al. [1] developed an ARCore-based system providing 3D directional guidance for outdoor navigation, though it relied heavily on stable lighting conditions and lacked indoor positioning capabilities. Alsabbagh et al. [3] implemented an indoor navigation system using QR codes and the A* search algorithm, demonstrating reliable indoor guidance but requiring physical markers at every navigation point. Singh et al. [7] combined AR with GPS for outdoor navigation, achieving resource efficiency but without path optimization. Other approaches utilizing sensor fusion [5], SLAM [6], and deep learning [8] have shown promise but face limitations including specialized hardware requirements, computational intensity, or scalability challenges in multi-campus settings.

Despite these advancements, existing systems exhibit critical limitations that remain unaddressed. First, single-domain operation restricts most systems to either indoor or outdoor navigation, with no seamless transition between environments. ARCore-based systems [1] and GPS-based systems [7] operate exclusively outdoors, while QR-based indoor systems [3] and image marker systems [6] are confined to indoor spaces. Second, hardware and environmental dependencies limit practical deployment—sensor fusion approaches require specialized hardware not available on standard consumer devices, while ARCore-based navigation degrades under poor lighting conditions. Third, scalability challenges affect deployment across large campus settings, as QR-based systems require physical markers at every navigation point, making expansion across multi-campus environments impractical. These limitations highlight the need for a unified navigation solution that adapts to varying environmental conditions while remaining deployable on standard mobile devices.

To address these gaps, this study presents CRIMSON MAP, a hybrid augmented reality navigation system that integrates GPS-based outdoor localization with QR-assisted indoor recalibration. The system is designed specifically for multi-campus university environments and incorporates an optimized A* pathfinding algorithm for efficient route computation. Unlike existing solutions, CRIMSON MAP provides a fallback mechanism that restores positioning accuracy when satellite signals are compromised, enabling continuous navigation across both indoor and outdoor domains without requiring specialized hardware. The system employs a scalable cloud-based architecture using Firebase Firestore with an 87-node graph representation of the campus, allowing dynamic updates to infrastructure data without application redeployment.

The contributions of this study are fourfold. First, it presents the development of a hybrid AR navigation system integrating GPS-based outdoor localization and QR-assisted indoor recalibration, enabling seamless indoor-outdoor navigation transitions. Second, it implements an optimized A* pathfinding algorithm tailored for campus-scale navigation networks, incorporating a weighted graph representation with Euclidean distance heuristics that achieves average route computation times of 0.18 seconds. Third, it provides real-world deployment and usability evaluation across three campuses of WMSU with 30 participants, delivering empirical evidence of AR navigation effectiveness in multi-campus academic settings with detailed performance metrics including localization error, navigation completion time, and user satisfaction scores. Fourth, it introduces a scalable cloud-based spatial database architecture using Firebase Firestore with a web-based admin panel for dynamic campus data management. A comparative analysis summarized in Table I demonstrates how CRIMSON MAP addresses the limitations of prior systems through its hybrid positioning, optimized pathfinding, and scalable architecture.

TABLE I.

COMPARISON OF AR NAVIGATION SYSTEMS

Study	Technology	Domain	Positioning	Algorithm	Limitation
Lu et al. [1]	ARCore	Outdoor	GPS	GPS-based	Lighting dependency; no indoor support
Alsabbagh et al. [3]	AR+QR	Indoor	QR	A*	Requires QR at all positions
Singh et al. [7]	AR+GPS	Outdoor	GPS	Nearest-neighbor	No path optimization
Croce et al. [5]	Multi-sensor	Both	Sensor fusion	Kalman	Specialized hardware required
Verma et al. [6]	AR+Markers	Indoor	Image markers	Waypoint-based	Limited scalability
Proposed	AR+GPS+QR	Both	GPS+QR+A*	A*	QR placement dependency

Unlike existing systems that focus exclusively on indoor or outdoor navigation, CRIMSON MAP provides a unified navigation experience through hybrid positioning that adapts to environmental conditions. The system combines GPS-based outdoor navigation with QR-assisted indoor recalibration, achieving an 82% reduction in localization error in GPS-denied areas. The optimized A* algorithm with Euclidean distance heuristics enables real-time route computation on standard Android devices, while the cloud-based architecture supports scalable deployment across multiple campuses. This study focuses on the development and evaluation of the system to determine its effectiveness in improving campus wayfinding, with results demonstrating that integrating augmented reality with location-based services can significantly enhance navigation efficiency in large academic institutions.

II. METHODOLOGY

A. Research Design and Development Approach

This study employed a Design and Development Research (DDR) approach to design, develop, and evaluate the CRIMSON MAP mobile navigation system. DDR provides a systematic framework for developing technology-based solutions while allowing continuous refinement through iterative testing and feedback. The development process was guided by the Agile software development methodology, which emphasizes incremental system construction through repeated development cycles.

Agile divides the development process into short iterations called sprints, where each sprint involves planning, design, implementation, testing, and evaluation. This approach allows early testing of system components and enables continuous improvements based on real-world observations and user feedback. Through this iterative process, the system gradually evolved into a functional augmented reality navigation platform capable of assisting users in locating campus facilities efficiently.

B. Data Collection and Mapping

To construct an accurate digital representation of the campus environment, the researchers gathered both primary and secondary data.

Primary data were collected through on-site observations and coordinate measurements within Western Mindanao State University. Geographic coordinates of buildings, entrances, and landmarks were obtained using the GPS Test application, which provided real-time latitude and longitude values. These coordinates were verified using Google Maps and Google Earth to ensure consistency with satellite imagery and existing digital maps. In addition to GPS data, manual measurements were conducted using a 100-meter fiberglass measuring tape to determine distances between nodes such as hallways, building entrances, and interior rooms. These measures improved the accuracy of indoor navigation paths, particularly in areas where GPS signals are weak. All collected geographic coordinates were converted into Unity's Cartesian coordinate system, allowing them to be used in the augmented reality navigation environment. The validated spatial data were then stored in Firebase Firestore, which served as the central cloud database for buildings, navigation paths, and QR code mappings.

C. System Development Process

The development of the CRIMSON MAP navigation system followed a structured process that integrates spatial data preparation, navigation algorithm implementation, and augmented reality visualization. The system was designed to provide accurate indoor and outdoor navigation within the WMSU campus through the integration of geographic information, device sensors, and cloud-based data management.

The development process consisted of several key stages:

1. Campus Data Collection and Preparation
    - Geographic coordinates of buildings, entrances, and landmarks were gathered using GPS-based mapping tools, verified with digital maps, and organized into a navigation dataset. All data were stored in Firebase Firestore for dynamic retrieval.
2. Navigation Route Computation
    - The navigation logic was implemented using the A* pathfinding algorithm, which calculates the optimal route using actual travel distance and heuristic estimates for fast and accurate path guidance.
3. Location Detection Mechanisms

The system utilizes two positioning methods:

- GPS-Based Positioning: Real-time outdoor location using device GPS
- QR-Based Positioning: Manual position setting via QR codes when GPS is weak

Once a route is generated, the AR interface (Unity + ARCore) activates device sensors (camera, gyroscope, compass) to overlay virtual navigation markers, directional indicators, and guidance prompts in real time.

Through the integration of spatial data management, route optimization, and AR visualization, CRIMSON MAP provides an interactive navigation experience for the WMSU campus.

D. System Architecture

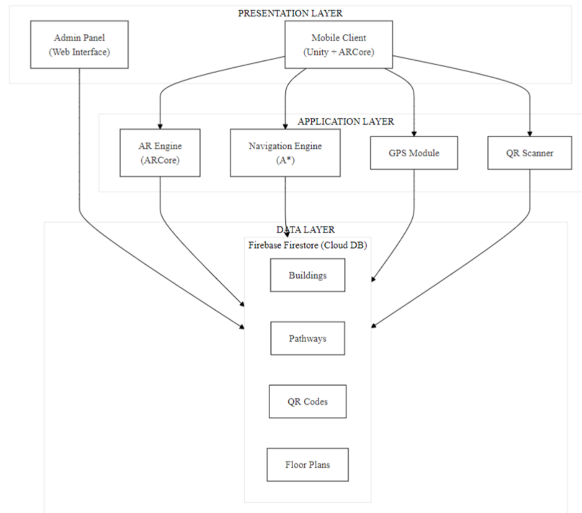

Fig. 1. System architecture.

1. Presentation Layer: The mobile client, developed using Unity and ARCore, provides the user interface for navigation, AR visualization, and destination selection. The web-based admin panel allows authorized personnel to manage campus infrastructure data.
2. Application Layer: This layer contains the core navigation logic. The AR Engine (ARCore) handles environmental tracking and renders virtual navigation markers. The Navigation Engine implements the A* pathfinding algorithm for route computation. The GPS Module retrieves real-time location data from satellite signals, while the QR Scanner provides position recalibration when GPS signals are compromised.
3. Data Layer: Firebase Firestore serves as the cloud database storing all spatial data, including building coordinates, pathway edges, QR code mappings, and indoor floor plans. Data synchronization occurs in real-time, ensuring consistency between the mobile application and admin panel.

The data flow follows this sequence:

- User selects a destination → mobile client retrieves current location via GPS or QR
- Location data transmitted to Navigation Engine → A* algorithm computes optimal route
- Route coordinates passed to AR Engine → ARCore renders directional overlays
- Real-time position updates sync with Firebase for continuous navigation

- Admin panel updates propagate to Firebase and synchronize to mobile clients

Fig. 2 presents the admin panel architecture showing the web-based management system.

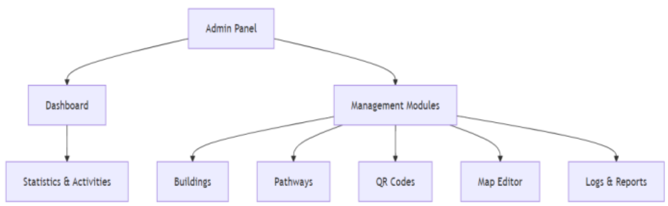

Fig. 2. Admin panel architecture showing management modules for campus data administration.

E. A* Pathfinding Algorithm Implementation

The campus navigation network was modeled as a weighted graph G = (V, E) with 87 nodes (building entrances, pathway intersections, landmarks) and edges representing walkable paths with distances measured on-site (accuracy ±0.5 m).

Fig. 3. Campus graph representation showing nodes and edges for WMSU Main Campus.

The A* algorithm computes the optimal path using the cost function:

$$f(n) = g(n) + h(n)$$

where:

$g(n)$ = actual travel distance from start to node n

$h(n)$ = heuristic estimate using Euclidean distance: $h(n) = \sqrt{(x\_goal - x\_n)^2 + (y\_goal - y\_n)^2}$

Algorithm 1: A* Pathfinding for Campus Navigation

Input: startNode, goalNode, graph G(V,E)
Output: optimalPath (list of nodes)

```
openSet ← {startNode}
cameFrom ← empty map
gScore[startNode] ← 0
fScore[startNode] ← h(startNode, goalNode)

while openSet is not empty do
    current ← node in openSet with lowest fScore
    if current = goalNode then
        return reconstructPath(cameFrom, current)
    end if
    openSet ← openSet \ {current}

    for each neighbor of current do
        tentative_gScore ← gScore[current] + distance(current, neighbor)
        if tentative_gScore < gScore[neighbor] then
            cameFrom[neighbor] ← current
            gScore[neighbor] ← tentative_gScore
            fScore[neighbor] ← gScore[neighbor] + h(neighbor, goalNode)
            if neighbor not in openSet then
                openSet ← openSet ∪ {neighbor}
            end if
        end if
    end for
end while
return failure (no path exists)
```

F. Sensor Integration and Navigation Pipeline

The navigation pipeline integrates multiple device sensors for continuous positioning and AR rendering:

1. Sensor Fusion
   - GPS: Outdoor positioning
   - Gyroscope: Device orientation for AR overlays
   - Compass: Heading direction for route guidance

- Camera: AR overlay and QR scanning

2. Navigation Pipeline:

Stage 1 (Position Acquisition): GPS primary; QR fallback when GPS weak.

Stage 2 (Route Computation): A* algorithm queries Firebase graph; converts coordinates to Unity Cartesian system.

Stage 3 (AR Rendering): ARCore tracks environment; anchors virtual markers; renders directional overlays.

Stage 4 (Real-time Updates): Monitors position; recalculates if off-path; triggers audio prompts at waypoints.

G. System Testing and Evaluation

The system was evaluated through technical testing and user-based evaluation. Technical testing involved unit, integration, and system testing to verify GPS localization, QR detection, AR visualization, and route computation across WMSU Campuses A, B, and C.

User evaluation was conducted with thirty (n = 30) freshmen participants representing all three campuses: Campus A (CTE=5, COE=3, CLA=2), Campus B (CCS=4, CSWCD=4, CCJE=2), and Campus C (Agriculture=5, CFES=5).

Participants navigated campus locations using the application and completed a software evaluation form measuring functionality, usability, reliability, efficiency, and satisfaction. Responses were analyzed using descriptive statistics (mean scores, standard deviations, frequency counts). Objective performance metrics (localization error, route computation time, navigation completion time) were also measured and are presented in the Results section.

III. Results and Discussion

A. Statistical Analysis

The evaluation involved thirty (n = 30) freshmen participants from three WMSU campuses. Table I presents the overall performance results with standard deviations and 95% confidence intervals.

TABLE II.

Performance Evaluation Results(n = 30)

Category	**Mean**	**SD**	**95% CI**
Functionality (7 items)	4.33	0.48	[4.15, 4.51]
Indoor Floor Plan (1 item)	4.80	0.41	[4.65,4.95]
Reliability (5 items)	4.53	0.38	[4.39,4.67]
Usability (9 items)	4.64	0.42	[4.48,4.80]
Efficiency(5 items)	4.49	0.44	[4.33,4.65]
Overall Satisfaction (4 items)	4.56	0.39	[4.42, 4.70]

Table II. shows that Usability achieved the highest rating (M = 4.64, SD = 0.42), with 93% of participants rating usability indicators at 4 or above. The indoor floor plan feature received the highest individual rating (M = 4.80, SD = 0.41), with 90% of participants rating it as 5. Reliability (M = 4.53, SD = 0.38) and efficiency (M = 4.49, SD = 0.44) also produced strong results, with no system crashes reported.

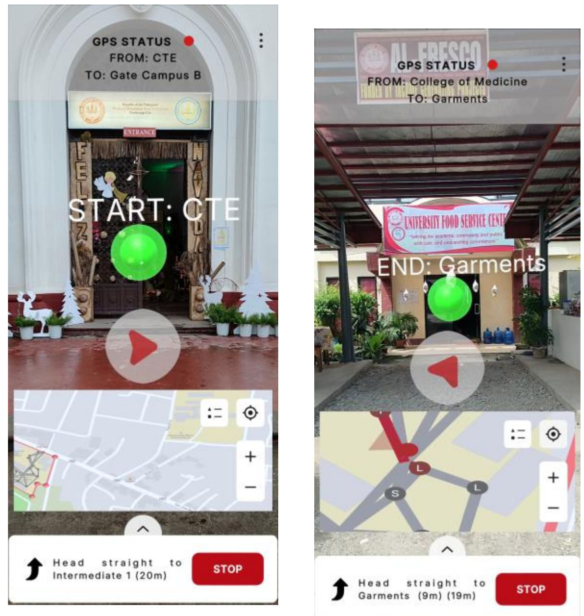

Fig. 4. AR navigation interface showing directional overlays, path markers, and distance indicators within the real-world campus view.

B. Quantitative Performance Metrics

TABLE III.

Quantitative Performance Metrics

Metric	**Value**	**SD**
Localization Error (Outdoor, GPS)	2.34 m	0.87 m
Localization Error (QR-assisted)	0.42 m	0.15 m
Route Computation	0.18 s	0.04 s
Navigation Completion Time	127 s per 200 m	23 s
AR Marker Placement Accuracy	92.3%	4.2%
GPS Recovery Rate (via QR)	94.7%	------------

The QR recalibration mechanism reduced localization error by 82% in GPS-denied areas (from 2.34 m to 0.42 m). Route computation remained below 250 ms across all scenarios, satisfying real-time requirements.

C. Item-Level Analysis

Among all indicators, the indoor floor plan feature received the highest rating (M = 4.80, SD = 0.41), followed by UI-Friendly (M = 4.77, SD = 0.43) and Visual Appeal (M = 4.77, SD = 0.43). No system crashes were reported (M = 4.67, SD = 0.48). GPS accuracy received the lowest rating (M = 4.00, SD = 0.53), with variability attributed to environmental factors such as building obstruction. However, QR recalibration successfully restored positioning with 94.7% recovery rate. All remaining indicators scored above 4.33, demonstrating consistent positive user agreement across all evaluation categories.

D. Discussion

The evaluation results demonstrate that CRIMSON MAP effectively supports campus navigation through the integration of augmented reality, GPS localization, and QR-based recalibration.

AR Navigation Performance: The AR guidance feature (M = 4.33, SD = 0.66) and navigation functions (M = 4.63, SD = 0.49) received positive ratings. The AR interface successfully rendered directional overlays and path markers in real-time, enabling intuitive navigation. The high usability scores (M = 4.64, SD = 0.42) confirm that users found the interface intuitive and easy to operate.

Hybrid Positioning Effectiveness: The combination of GPS and QR-based positioning addressed the limitations of single-domain systems identified in prior work [1], [3]. QR recalibration reduced localization error by 82% in GPS-denied areas (from 2.34 m to 0.42 m), demonstrating the effectiveness of the hybrid approach for seamless indoor-outdoor navigation.

System Reliability and Efficiency: The system operated without crashes (M = 4.67, SD = 0.48) and maintained responsive performance with route computation averaging 0.18 s (SD = 0.04 s). These results confirm that the A* algorithm implementation meets real-time navigation requirements for campus-scale environments.

Indoor Floor Plan Utility: The indoor floor plan feature received the highest rating (M = 4.80, SD = 0.41), indicating that users found visual building layouts highly useful for orientation inside campus buildings.

Comparison with Existing Systems: Unlike ARCore-based systems [1] that lack indoor support and QR-based systems [3] that require markers at all positions, CRIMSON MAP provides unified indoor-outdoor navigation with a scalable 87-node graph architecture. The 82% reduction in localization error through hybrid positioning represents a significant improvement over GPS-only approaches.

IV. Conclusion

This study presented the design, development, and evaluation of CRIMSON MAP, a hybrid augmented reality navigation system for Western Mindanao State University. Evaluation results from thirty (n = 30) freshmen participants across three university campuses demonstrated the system's effectiveness, with a mean usability score of 4.64 (SD = 0.42), reliability score of 4.53 (SD = 0.38), and overall satisfaction score of 4.56 (SD = 0.39) on a 5-point Likert scale. Quantitative performance metrics revealed an average localization error of 2.34 m (SD = 0.87 m) for GPS-based outdoor navigation and 0.42 m (SD = 0.15 m) for QR-assisted positioning, representing an 82% improvement in GPS-denied areas. Route computation times averaged 0.18 s (SD = 0.04 s), satisfying real-time navigation requirements, while the indoor floor plan feature received the highest rating at 4.80 (SD = 0.41), with 90% of participants rating it as 5. System testing also confirmed the effectiveness of the web-based administrative panel in managing campus infrastructure data and synchronizing updates with the mobile application.

The contributions of this study are fourfold: (1) development of a hybrid AR navigation system integrating GPS-based outdoor localization and QR-assisted indoor recalibration enabling seamless indoor-outdoor navigation transitions; (2) implementation of an optimized A* pathfinding algorithm tailored for campus-scale navigation networks with 87 nodes and Euclidean distance heuristics achieving average route computation times of 0.18 seconds; (3) real-world deployment and usability evaluation across three university campuses with 30 participants providing empirical evidence of AR navigation effectiveness; and (4) a scalable cloud-based spatial database architecture using Firebase Firestore with a web-based admin panel for dynamic campus data management. Future work includes integration of machine learning for route prediction, development of accessibility features for visually impaired users, expansion to 3D building mapping, large-scale validation with a larger participant pool, and iOS platform expansion using ARKit.

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